Quantization in Digital Image Processing

Question: A grayscale image is converted into a digital image. Describe the role of quantization in this process and its impact on image detail.

Introduction

A digital image is formed through two main processes: **sampling** and **quantization**. Sampling converts the continuous image into discrete pixels, while quantization converts the continuous pixel intensity values into a finite number of gray levels.

Role of Quantization

Quantization assigns each pixel intensity to the nearest available gray level. For an 8-bit image there are 256 gray levels (0–255). Reducing the number of gray levels decreases the intensity resolution of the image.

Impact on Image Detail

Higher quantization levels (64 or 256 levels) preserve smooth intensity transitions and image details. Lower quantization levels (16, 8, 4, or 2 levels) cause loss of fine details, visible banding, posterization, and reduced contrast. Therefore, choosing an appropriate quantization level is important for maintaining image quality while reducing storage requirements.

Code Explanation

The program creates a grayscale gradient image and quantizes it into different gray-level resolutions (64, 32, 16, 8, 4, and 2 levels). The displayed images demonstrate how reducing the number of intensity levels gradually decreases image quality.

Python Code

```
Include the complete Python code provided in the question here.
```

Result

The output clearly shows that as the number of quantization levels decreases, the image loses smooth gray-level transitions and appears more segmented. This illustrates the effect of quantization on digital image quality.